15 (a) Show that the equation $x^3 + 7x - 5 = 0$ has a solution between $x = 0$ and $x = 1$

Let $f(x) = x^3 + 7x - 5$.

$f(0) = -5$; $f(1) = 1 + 7 - 5 = 3$.

1 mark
Evaluate at both endpoints

f is continuous and changes sign, so a solution lies between 0 and 1.

1 mark
Correct values and the sign-change conclusion

(2)

(b) Show that the equation $x^3 + 7x - 5 = 0$ can be arranged to give $x = \dfrac{5}{x^2 + 7}$

Add 5 to both sides: $x^3 + 7x = 5$

1 mark
Correct first rearrangement

×	x^2	7
x	x^3	$7x$

$x(x^2 + 7) = 5$

$\div(x^2 + 7)$ $\quad$ $\div(x^2 + 7)$

1 mark
Factorise, then divide both sides by $x^2 + 7$ (always positive)

$$x = \frac{5}{x^2 + 7}$$

(2)

(c) Starting with $x_0 = 1$, use the iteration formula $x_{n+1} = \dfrac{5}{x_n^2 + 7}$ three times to find an estimate for the solution of $x^3 + 7x - 5 = 0$

$x_1 = 5 \div (1^2 + 7) = 0.625$

1 mark
First substitution uses $x_0 = 1$

$x_2 = 5 \div (0.625^2 + 7) = 0.6765327696...$

$x_3 = 5 \div (x_2^2 + 7) = 0.6704483001...$

1 mark
Use each unrounded result in the next substitution

After exactly three iterations: $x \approx 0.6704483001$.

1 mark
Correct third iterate

0.6704483001

(3)

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(d) By substituting your answer to part (c) into $x^3 + 7x - 5$,
comment on the accuracy of your estimate for the solution to $x^3 + 7x - 5 = 0$

Using the unrounded third iterate:

$x_3^3 + 7x_3 - 5 = -0.0054947698...$

1 mark
Substitute into the original expression

This is close to 0, so the estimate is close to a solution.

1 mark
Comment on the small residual, rather than claiming an exact root

(2)

(Total for Question 15 is 9 marks)

16 The petrol consumption of a car, in litres per 100 kilometres, is given by the formula

$$\text{Petrol consumption} = \frac{100 \times \text{Number of litres of petrol used}}{\text{Number of kilometres travelled}}$$

Nathan's car travelled 148 kilometres, correct to 3 significant figures.
The car used 11.8 litres of petrol, correct to 3 significant figures.

Nathan says,

"My car used less than 8 litres of petrol per 100 kilometres."

Could Nathan be wrong?
You must show how you get your answer.

(Total for Question 16 is 3 marks)

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